CSD361 — Lab Assignment 2

Classification with Logistic Regression and Linear SVM:
Model Building, Performance Metrics, and Comparative Evaluation

Deadline: 22 October 2026, 11:59 PM.

Objective

To build, evaluate, and compare two linear classifiers — Logistic Regression and a Linear Support Vector Machine — on a real medical dataset, to see where a linear model fails and how feature expansion helps, and to understand the classification-evaluation metrics (confusion matrix, precision, recall, ROC curve, ROC-AUC) well enough to justify a model choice from experimental evidence. Each section mixes conceptual understanding, implementation, and interpretation.

Dataset

- **Primary dataset:** Breast Cancer Wisconsin (Diagnostic) — WDBC, via scikit-learn sklearn.datasets.load_breast_cancer (originally UCI ML Repository). — https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic
- **Size:** 569 tumours, 30 real-valued features. **Classes:** 212 malignant, 357 benign.
- **Positive class (fixed):** malignant = 1, benign = 0. Note: scikit-learn codes malignant = 0, so you must relabel (Section 1).
- **One sub-part (2.5) uses a small synthetic dataset** (two concentric circles, via sklearn.datasets.make_circles) purely to illustrate a limitation of linear models — this does not replace WDBC.

Submission Guidelines:

1. **Read each question carefully** and provide a clear, brief, and well-structured solution.
2. For programming or implementation-based questions :
    a. Write readable and well-commented code, and an output screenshot is required.
    b. Clearly mention the dataset, preprocessing steps, algorithms, and parameters used.
    c. Include relevant output, graphs, or performance measures where required.
3. Prepare the documentation with all code screenshots and output screenshots (wherever applicable) and upload a solution PDF.
4. Also, provide the link to your code file in the document itself (for example, Jupyter Notebook, lab, or Colab), whatever you are using. We will be using the same code file during the viva for this assignment.
5. Use Python 3 with NumPy, pandas, Matplotlib and scikit-learn. Set random seed = 42 everywhere.
6. Use ONE split for both models on WDBC: 80/20, stratified, random_state = 42 (same test examples for both).
7. Standardise features with StandardScaler fitted on the training set only, then transform train and test. (A Pipeline is an optional convenience, not required.)
8. For ROC/AUC use continuous scores: predict_proba(...)[:,1] for Logistic Regression and decision_function(...) for the SVM. Do not treat SVM scores as probabilities.
9. Every plot needs a title, axis labels, and legend; every confusion matrix must label the Actual and Predicted axes. Report only results your own code produced.

10. The deadline for the submission is 22 October 2026, 11:59 PM.

11. **Please note that this assignment contains 50 marks, and 50 marks will be for the related VIVA. Total -100 marks.**

12. **The viva will be conducted on Friday, 23 October 2026, for Assignment 2 during Lab hours.**

Section 1 [5 marks] — Dataset Exploration (Each question carries 1 mark)

1.1 Load the WDBC dataset and display its shape (number of samples and features), the first few feature names, and the target names.

1.2 Show the class distribution (counts of malignant vs benign). Relabel the target so that malignant = 1 and benign = 0, and confirm the positive-class count is 212.

1.3 Split into train and test using an 80/20 stratified split (random_state = 42). Report the number of training and test samples and the number of positives in the test set, and explain why the test set must be kept unseen during training and tuning.

1.4 Explain: (a) why feature scaling matters for BOTH Logistic Regression and a Linear SVM; (b) why the scaler must be fitted on the training data only; and (c) how fitting the scaler on the complete dataset before splitting causes data leakage and makes the reported test performance over-optimistic.

1.5 State which class is positive and which is negative here, and explain why a false negative (malignant predicted benign) and a false positive (benign predicted malignant) carry different real-world costs.

Section 2 [15 marks]— Logistic Regression

2.1 **[2 marks]** Logistic Regression is used for classification even though its name contains "regression". Explain why, state what the sigmoid output represents, and explain how a threshold converts that output into a predicted class.

2.2 **[2 marks]** Train a Logistic Regression model on the standardised WDBC training set. State the key configuration (e.g., max_iter) and the role of the sigmoid. Then, on the test set, obtain the predicted labels and the positive-class probabilities, and display the confusion matrix with labelled axes. Report the training-set and test-set accuracy.

2.3 **[3 marks]** Report precision and recall for the positive (malignant) class, showing how at least one follows from the confusion-matrix counts; then compute the ROC-AUC from the positive-class probabilities and interpret it in one or two sentences. (The ROC curve itself is drawn in Section 4.)

2.4 **[3 marks]** First explain how raising or lowering the classification threshold changes false positives and false negatives, and give one application where you would deliberately LOWER the threshold. Then, for the 10 samples below (actual label; predicted positive-class probability):

Sample	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Actual	1	1	0	1	0	1	0	0	1	0
P (positive)	0.95	0.80	0.70	0.60	0.55	0.45	0.40	0.30	0.20	0.10

At thresholds t = 0.50 and t = 0.30, give the predicted labels, the confusion matrix (TP, FP, TN, FN), and precision and recall, and state what happened to precision and recall as the threshold was lowered, and why.

2.5 **[5 marks]** (Non-linear data experiment) Generate a dataset that is NOT linearly separable using sklearn.datasets.make_circles (two concentric circles; use some noise and a fixed random_state). (i) Train Logistic Regression on the two raw features (x1, x2) and report its test accuracy — you should observe it fails (close to chance). (ii) Now add the polynomial features $x1^2$, $x2^2$, and $x1 \cdot x2$ to the inputs and train Logistic Regression again; report the new test accuracy and show that it improves substantially. (iii) Explain WHY the polynomial features fix the problem (what shape the boundary $x1^2 + x2^2$ = constant describes, and how expanding the feature space makes the two classes linearly separable).

Section 3 [15 marks] — Support Vector Machine

3.1 **[2 marks]** Define briefly with meaning: (i) SVM Hyperplanes, (ii) Margin, (iii) Support Vectors. Then explain why an SVM prefers the maximum-margin boundary over any other decision boundary.

3.2 **[3 marks]** In one to two sentences, explain why feature scaling matters for an SVM. Then train a LINEAR SVM (linear kernel, C = 1.0) on the standardised WDBC training set; report its training-set and test-set accuracy; and on the test set obtain the predicted labels and the decision_function scores and display the confusion matrix with labelled axes (note in one line why the decision scores are NOT probabilities).

3.3 **[3 marks]** Report precision and recall for the positive class, then compute the ROC-AUC from the decision scores and interpret it in one sentence. (The ROC curve itself is drawn in Section 4.)

3.4 **[3 marks]** (A) Plot and explain briefly how making C much larger would be expected to affect the margin width and the tendency to overfit (comment only on the margin/overfitting trade-off; do NOT assume the number of support vectors changes by any fixed amount).

(B) **[4 marks]** The SVM uses hinge loss while Logistic Regression uses log-loss: show with the trained model and describe how these two training objectives differ, including what hinge loss penalises and when it is exactly zero.

Section 4 [15 marks] — Performance Metric Comparison

4.1 **[3 marks]** Using scikit-learn on the test set, obtained from BOTH trained models (above) — Logistic Regression and the Linear SVM — the confusion matrix and the precision, recall, accuracy, and F1-score. State the formula for each metric (e.g. precision = TP/(TP+FP), F1 = 2·P·R/(P+R)), and explain what precision and recall mean for each model in this medical setting — i.e. what a false positive and a false negative represent here.

4.2 **[3 marks]** Assemble a single side-by-side comparison table for your two trained models (same test set) containing TP, FP, TN, FN, precision, recall, accuracy and ROC-AUC, taken from Sections 2 and 3.

Comparison table template to fill (use your own results):

Metric	Logistic Regression	Linear SVM
TP / FP / TN / FN	— / — / — / —	— / — / — / —
Precision	—	—
Recall	—	—
Accuracy	—	—
ROC-AUC	—	—

4.3 **[3 marks]** (Threshold & ROC behaviour of your trained model) Using your Logistic Regression's predicted positive-class probabilities on the test set, change the decision threshold to 0.30, 0.50 and 0.70; for EACH threshold recompute the confusion matrix, precision and recall, and record the corresponding (FPR, TPR) operating point. Put these in a small table. Then plot a single figure with BOTH trained models' complete ROC curves

plus the diagonal random-guess line (axis labels and a legend showing each AUC), and explain in two or three sentences how varying the threshold moves the operating point along the ROC curve.

4.4 **[3 marks]** Distinguish (i) what the confusion matrix, precision, and recall tell you at the default 0.5 threshold (ii) what the ROC curve and AUC tell you across all thresholds. Then state which of YOUR two models you would deploy for first-pass cancer screening and why, in terms of the cost of a false negative; and name one kind of application where the other model's higher precision (fewer false positives) would instead be preferable.

4.5 **[3 marks]** (Using your own results) Answer briefly:

(a) Does one of your two models have a higher ROC-AUC yet a lower recall at the 0.5 threshold? — explain briefly why it happens.

(b) Both models reach about x% accuracy — why is accuracy alone an insufficient basis for choosing between them here, and which of your other metrics would you rely on?

(c) Using the training-vs-test accuracy you reported for each model, what would a large gap indicate about overfitting, and how does this relate to the SVM's parameter C?